ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION â€" DECEMBER / JUNE 2024

B.TECH. CSE ENGINEERING â€" SEMESTER 3

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted. Mobile phones are strictly prohibited.

--- UNIT I ---

Q.1 (a) Explain the asymptotic notations (Big-O, Omega, Theta) used for algorithm analysis with mathematical definitions. [7 Marks]
(b) Write an algorithm to insert and delete an element from a Circular Queue using an array. Discuss the boundary conditions. [7 Marks]

[OR]

Q.1 (Alt) Alternative: (a) Compare Singly Linked List, Doubly Linked List, and Circular Linked List in terms of time and space complexity. [7 Marks]
(b) Write a C/C++ program to reverse a singly linked list in-place using iterative approach. [7 Marks]

--- UNIT II ---

Q.2 (a) What is Stack? Explain how infix expression is converted to postfix expression with an example: ((A + B) * C - (D - E)) ^ (F + G). [7 Marks]
(b) Describe the Tower of Hanoi problem. Write a recursive algorithm and derive its recurrence relation and time complexity. [7 Marks]

[OR]

Q.2 (Alt) Alternative: (a) Implement a Double-Ended Queue (Deque) using linked list with insert_front and delete_rear operations. [7 Marks]
(b) What are Priority Queues? Explain array-based and heap-based representations. [7 Marks]

--- UNIT III ---

Q.3 (a) Define Binary Search Tree (BST). Construct a BST for the following keys: 45, 23, 67, 12, 34, 89, 56, 78, 90. Show inorder traversal. [7 Marks]
(b) What is an AVL tree? Explain LL, RR, LR, and RL rotations with suitable diagrams. [7 Marks]

[OR]

Q.3 (Alt) Alternative: (a) What are B-Trees? Explain insertion of elements in a B-Tree of order 3. [7 Marks]
(b) Explain Huffman Coding algorithm. Construct Huffman tree for characters A, B, C, D, E with frequencies 15, 9, 12, 18, 16. [7 Marks]

--- UNIT IV ---

Q.4 (a) Differentiate between BFS and DFS graph traversals with pseudocode and time complexities. [7 Marks]
(b) Apply Dijkstra's Single Source Shortest Path algorithm on a directed weighted graph. [7 Marks]

[OR]

Q.4 (Alt) Alternative: (a) Explain Prim's and Kruskal's algorithms for finding Minimum Spanning Tree with an example. [7 Marks]
(b) What is Topological Sorting? Find topological order for a given Directed Acyclic Graph (DAG). [7 Marks]

--- UNIT V ---

Q.5 (a) Explain Quick Sort algorithm with partition strategy. Analyze best, worst, and average case time complexities. [7 Marks]
(b) What is Hashing? Discuss Collision Resolution Techniques: Linear Probing, Quadratic Probing, and Separate Chaining. [7 Marks]

[OR]

Q.5 (Alt) Alternative: (a) Explain Merge Sort algorithm with recursion tree. Prove that its time complexity is O(N log N). [7 Marks]
(b) Describe Heap Sort algorithm. Build a Max-Heap for: 4, 10, 3, 5, 1, 9, 8 and show sorting steps. [7 Marks]

**** END OF QUESTION PAPER â€" MITS GWALIOR AUTONOMOUS EXAMINATION SYSTEM ****